BEING A UNIVERSAL HOMEOMORPHISM CAN BE CHECKED AFTER A FLAT COVER OF THE TARGET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The reduction of universal injectivity to a cover of the target by an algebraic space. Invoked at the start of the counted proof (source lines 3185-3187) as a pointer to a lemma of this chapter. Transfers universal injectivity from the base change to the morphism itself.
- The openness on topological spaces of surjective flat morphisms of finite presentation, and the formation of the topological space of an algebraic stack as a quotient of the space of a smooth cover. Used inside the counted proof (source lines 3190-3211). Turns the homeomorphism on the cover into openness of the induced map on the topological spaces of the test stacks.

Source reference. Original lemma lemma-check-universally-injective-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3187–3187. Navigation pointer only: the counted text refers to the Stacks item lemma-check-universally-injective-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $W \to \mathcal{Y}$ be surjective, flat, and locally of finite presentation where W is an algebraic space. If the base change $W \times_{\mathcal{Y}} \mathcal{X} \to W$ is a universal homeomorphism, then f is a universal homeomorphism.*

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Proof. Assume $g : W \times_{\mathcal{Y}} \mathcal{X} \to W$ is a universal homeomorphism. Then g is universally injective, hence f is universally injective by Lemma lemma-check-universally-injective-covering. On the other hand, let $\mathcal{Z} \to \mathcal{Y}$ be a morphism with $\mathcal{Z}$ an algebraic stack. Choose a scheme U and a surjective smooth morphism $U \to W \times_{\mathcal{Y}} \mathcal{Z}$. Consider the diagram

$$\begin{array}{ccccc} W \times_{\mathcal{Y}} \mathcal{X} & \longleftarrow & U \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & \mathcal{Z} \times_{\mathcal{Y}} \mathcal{X} \\ \downarrow{\scriptstyle g} & & \downarrow & & \downarrow \\ W & \longleftarrow & U & \longrightarrow & \mathcal{Z} \end{array}$$

The middle vertical arrow induces a homeomorphism on topological space by assumption on g. The morphism $U \to \mathcal{Z}$ and $U \times_{\mathcal{Y}} \mathcal{X} \to \mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}$ are surjective, flat, and locally of finite presentation hence induce open maps on topological spaces. We conclude that $|\mathcal{Z} \times_{\mathcal{Y}} \mathcal{X}| \to |\mathcal{Z}|$ is open. Surjectivity is easy to prove; we omit the proof. $\square$